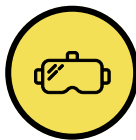




Immersive Technologies



Scope

Digital technologies shift collaboration, communication, and social presence into fully virtual or mixed reality workspaces. The focus is on the shared interaction space, avatars, and the experience of organizational belonging.

Situation

Teams predominantly work, communicate, and make decisions in virtual spaces using avatars. While some employees thrive, others experience a loss of belonging, social proximity, and organizational identity. In the virtual room, information can be easily translated and intentionally shown or hidden.

Leadership Impact

Leadership must establish trust, cohesion, and orientation without reliable physical presence and determine which interactions should take place immersively. At the same time, alternatives are necessary for employees who feel disconnected.



Personal AI Agents



Scope

Personal or role specific AI representatives communicate, coordinate, negotiate, or prepare decisions on behalf of identifiable individuals. The focus is on delegation, representation, and the relationship between human command and machine action.

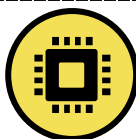
Situation

Leaders and teams are increasingly delegating communication, scheduling, negotiations, and operational decisions to AI agents. For others, it is not always apparent whether a task is currently authorized, whether a person has intervened, or whether two digital representatives have made a binding agreement with one another.

Leadership Impact

Leadership must define the limits of delegation, disclosure requirements, and escalation paths. They must ensure that speed does not lead to relationships, negotiation skills, and human accountability creeping out of the organization.

Ubiquitous Sensor Technology



Scope

A comprehensive sensor infrastructure continuously collects behavioral, performance, environmental, and health data from people, machines, and spaces. The focus is on data collection, networking, and transparency before assessments are derived from the data.

Situation

Sensors in clothing, buildings, vehicles, and work equipment continuously generate real time data about the performance, behavior, and condition of employees. This transparency supports safety, maintenance, and resource management, but makes it difficult for employees to evade measurement.

Leadership Impact

Leadership must limit the purpose, scope, storage duration, and access rights and create genuine choices. They must decide what transparency is necessary for operations and at what point technically possible measurement disproportionately impairs trust, privacy, and autonomy.

Generative AI 2.0



Scope

AI systems generate texts, images, software, models, simulations, and strategic designs at a higher quality than humans. The focus is on the generation and curation of knowledge and creative outputs, as well as the transformation of human expertise.

Situation

Core knowledge and creative tasks are performed together with or largely by AI. Professionals are evolving from producers to curators and individuals responsible for AI outputs. As quality increases, it becomes harder to determine whether/where human competencies should be added to the process.

Leadership Impact

Leadership must redefine role models, quality standards, and learning paths. They must clarify where human expertise should remain visible and verifiable, how the loss of competence is avoided, and who is responsible for content whose creation can hardly be fully traced.



Autonomous Robots



Scope

Physical autonomous systems that perceive and act within real-world environments, coordinate with humans and other machines, and make context-dependent decisions without continuous human control. The focus is on physical agency, human-robot collaboration, intervention rights, and accountability for autonomous actions.

Situation

Autonomous robots perform cross-functional tasks in production, logistics, maintenance, care, and security, and respond to their environment in real time. Employees are increasingly working in sync with and under the direction of machines whose decisions they cannot fully understand or stop in time in critical situations.

Leadership Impact

Leadership must define the rights to intervene, safe shutdown procedures, liability, and the division of labor between humans and machines. It must decide whether employees remain formally responsible, even though they may not have sufficient insight or control.



Brain-Computer Interfaces



Scope

Invasive neurotechnological interfaces convert neural activity into a digital data stream that connects brains to devices, platforms, and other BCI users. This allows information, intentions, and system responses to be transmitted without conventional physical or verbal input.

Situation

Neural data flows directly into work processes, digital platforms, and communication among connected employees. This creates new dependencies on technical infrastructures, highly sensitive mental data trails, and a growing disparity between employees with and without access to these interfaces.

Leadership Impact

Leadership must define the limits of engagement, the voluntary nature of participation, equivalent alternatives, and data protection rights for mental data. It must also be clarified who bears responsibility when an action arises from a combination of neural signals, system interpretation, and automatic responses.

Collective Memory Systems



Scope

Systems that comprehensively and automatically store decision paths, justifications, outcomes, and organizational experiences. The data is automatically linked and processed to be permanently available for both humans and AI agents. The focus is on organizational memory, traceability, and conscious forgetting.

Situation

Organizations systematically refer to past decisions, exceptions, and behavioral patterns. This makes actions more traceable, and knowledge remains available despite staff turnover. At the same time, old assumptions, errors, and power dynamics can repeatedly flow into new recommendations.

Leadership Impact

Leadership must decide what may be stored, corrected, or consciously forgotten. They must strike a balance between traceability and renewal, preventing organizational memory from becoming an unquestionable authority.

Behavioral Threat Detection



Scope

Systems that use multimodal data to draw conclusions about safety-related, psychological, or non-compliant behavioral changes among employees. Key issues here include risk inference, prevention, false positives, and the challengeability of algorithmic assessments.

Situation

Models identify potential overload, fraud, radicalization, insider risks, or security breaches before a specific incident occurs. This enables earlier prevention, but it can also lead to false alarms, cultural biases, and persistent suspicion profiles.

Leadership Impact

Leadership must strike a balance between care, prevention, and control, and set high standards for evidence, purpose limitation, and challengeability. Those affected must not be disadvantaged solely on the basis of behavioral patterns that are difficult to explain.



Cognitive Pharmacology



Scope

Targeted, time-limited, and generally reversible pharmacological interventions designed to influence concentration, learning ability, alertness, or stress resilience. The immediate effects, health risks, and work-related performance expectations are taken into account.

Situation

Substances designed to enhance cognitive performance are becoming increasingly common in particularly demanding professional settings. The lines between personal choice, medical support or prevention, and implicit expectations become blurred as soon as optimized performance becomes the new benchmark.

Leadership Impact

Leadership must determine whether and to what extent such interventions are acceptable in the workplace. Health and safety, confidentiality, and voluntary participation must still apply even when the technology is used privately, but performance expectations are collectively adjusted.



Neurofeedback at Scale



Scope

Wearable and neurotechnological systems track cognitive or emotional states and continuously provide this feedback to users to aid in self-regulation. Self-regulation, the ability to measure internal states, and the line between this and performance monitoring are available to every employee and, where applicable, their supervisors.

Situation

Employees receive real-time information about their concentration, stress, fatigue, or emotional regulation and adjust their behavior accordingly. This can facilitate self-regulation, but at the same time transforms internal states into measurable performance indicators that become indirectly comparable.

Leadership Impact

Management must prevent a voluntary self-assessment tool from becoming an implicit performance monitoring mechanism. Issues that need to be clarified include data access, the tool's validity, the right to opt out, and how to handle results that indicate stress without explaining its causes.

Hyper Personalized Incentive Systems



Scope

Dynamic digital systems adapt information, tasks, rewards, and decision architectures based on individual behavioral data. The focus is on personalized digital behavior steering, intransparent influence mechanisms, and the boundary between support and manipulation.

Situation

Information flows, goals, and incentives are personalized in real time to steer behavior in a targeted manner. Two employees receive different arguments or rewards for the same decision. This increases effectiveness and precision, while common rules and the boundary to manipulation become blurred.

Leadership Impact

Leadership must establish transparency, consent, and limits to permissible behavior steering. They must decide whether a measure still supports autonomy or already undermines it if the affected individuals cannot recognize the mechanism of influence or practically escape it.

Mixed Reality Filters



Scope

Augmented reality systems overlay the physical environment with layers of information that are individually prioritized, hidden, or rated. The focus is on the fragmented perception of the same situation and the question of establishing a binding, shared information base.

Situation

Employees see different pieces of information, risk assessments, and priorities during meetings, in production areas, or in customer interactions. Decisions are made faster and better, but a shared reality breaks down because it is unclear which information others have seen, not seen, or weighted differently.

Leadership Impact

Leadership must define a binding, shared information base, rights to shape filters, and transparent protocols. It must prevent personalized perceptions from distributing power unnoticed or making it impossible to reconstruct erroneous decisions in retrospect.



Operations Planning AI



Scope

AI systems autonomously forecast, coordinate, and continuously adjust organization-wide processes according to predefined objectives. They make operational decisions, monitor their effects, and adapt subsequent actions through closed feedback loops.

Situation

Staff allocation, pricing, procurement, production planning, and resource allocation are largely self-optimizing processes. Human intervention now occurs only at the meta-level. When goals conflict or models drift, decisions are made whose triggers no one can trace.

Leadership Impact

Leadership must define objectives for decision-making, as well as specify which decisions the system is authorized to make autonomously and when human intervention is required. System decisions must remain transparent, verifiable, and overridden at any time. Efficiency should be balanced with security and fairness.



Deepfakes



Scope

AI-generated or manipulated audio, video, document, and identity artifacts, as well as methods that make the origin and editing of digital content verifiable. Authenticity, verifiability, and trust in digital evidence are ongoing topics of discussion.

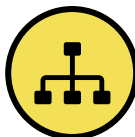
Situation

Deceptively real recordings, instructions, and identities circulate in business processes. At the same time, involved parties can reject genuine evidence as a forgery. Decisions slow down because authenticity can no longer be taken for granted and technical verification signals are themselves contested.

Leadership Impact

Leadership must establish binding verification paths, approvals, and response rules for doubtful content. They must remain capable of acting without leaving control over the organization to convincing forgeries or blanket mistrust.

Simulations and Strategic AI



Scope

Dynamic simulation models and predictive LLMs that map an organization's processes, markets, resources, and behavior generate possible futures and test the consequences of future decisions. Model assumptions and simulation results influence real-world decisions.

Situation

Before undertaking restructuring, making investments, or implementing crisis measures, the organization tests several scenarios using simulations or AI models. The results enable rapid decision-making, even though rare events, social and ethical values, and human reactions are only partially modeled.

Leadership Impact

Leadership must determine what decision-making authority simulations and AI tools are granted for strategic decisions, how assumptions are made transparent and models are validated, and when it is necessary to challenge an apparently unambiguous result. Human and societal values must not be overlooked.

Regenerative Medicine



Scope

High-cost regenerative therapies are funded by companies as exclusive health benefits. The treatment of chronic conditions results in long-term commitments between employers and employees.

Situation

In the competition for skilled workers, companies are increasingly offering expensive regenerative therapies. This raises new questions regarding financing, liability, and potential expectations regarding the loyalty and increased performance of the employees receiving treatment.

Leadership Impact

Leadership must assess whether subsidized health care services create indirect expectations of performance or become the standard for employability. At the same time, risks to motivation, ethical boundaries, liability, and follow-on costs must be addressed.



Human Enhancements



Scope

Biological, pharmacological, or neurotechnological interventions permanently enhance human abilities beyond what is therapeutically necessary. The focus is on performance enhancement, voluntary participation, fairness, and long-term physical and psychological consequences.

Situation

Companies fund initiatives designed to boost concentration, resilience, or learning ability. This results in benefits for more productive employees, but at the same time creates indirect pressure to adapt and new inequalities.

Leadership Impact

Management must define performance standards, voluntary participation, and protections against discrimination. It must decide whether fairness means the same rules for everyone, or whether employees who do not use performance-enhancing measures require special protection from structural pressure to adapt.